

Theory 13.8.6

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1 Aggregation by attention

The aggregation methods discussed so far either weight the contribution of the neighbors equally or in a way that depends on the graph topology. Conversely, in graph attention layers, the weights depend on the data at the nodes. A linear transform is applied to the current node embeddings so that:

$$\mathbf{H}'_k = \beta_k \mathbf{1}^T + \Omega_k \mathbf{H}.$$

Then the similarity s_{mn} of each transformed node embedding $\mathbf{h}'_m$ to the transformed node embedding $\mathbf{h}'_n$ is computed by concatenating the the pairs, taking a dot product with a column vector ϕ_k of learned parameters, and applying an activation function:

$$s_{mn} = a \left(\phi_k^T \begin{bmatrix} \mathbf{h}'_m \\ \mathbf{h}'_n \end{bmatrix} \right).$$